Week 10: 26–30 October 2026

الأسبوع 10: من 26 - إلى 30 أكتوبر 2026

Subject / الموضوع: AP Chemistry **Unit Title / عنوان الوحدة: Unit 7 Equilibrium – الاتزان الكيميائي** **Grade / الصف: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Session 1 الحصة الأولى 7.3 Reaction Quotient and Equilibrium Constant حاصل التفاعل وثابت الاتزان	**DOK 3 – Analyze:** I can write Q and K expressions for reversible reactions and analyze how Q relates to K, justifying which species are included in each expression.	3.3 Guide processing with explicit steps, chunking, and interactive models. 2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. **Formative:** writing Q and K expressions and mini-whiteboard checks	Kindly check Toddle for the lesson PPT and practice sheet.
Session 2 الحصة الثانية 7.4 Calculating the Equilibrium Constant حساب ثابت الاتزان	**DOK 3 – Analyze:** I can analyze experimental concentration or pressure data to calculate Kc or Kp, using an ICE table when only initial values and one equilibrium value are given.	5.3 Build fluency with graduated scaffolds and models. 6.2 Support planning with templates, strategy prompts, and modeled thinking. 8.3 Foster collaboration and community with structured group roles and supports. **Formative:** ICE-table practice problems and exit ticket	Kindly check Toddle for the lesson PPT and practice sheet.
Session 3 الحصة الثالثة 7.5 Magnitude of the Equilibrium Constant مقدار ثابت الاتزان	**DOK 3 – Explain:** I can analyze the magnitude of K to explain whether reactants or products are favored at equilibrium, and justify my predictions with particle-level reasoning.	3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 7.2 Make learning relevant, authentic, and connected to students' backgrounds. **Formative:** K-value ranking task and AP-style MCQs	Kindly check Toddle for the lesson PPT and practice sheet.
Session 4 الحصة الرابعة Unit 7 Checkpoint (College Board)	**DOK 3 – Apply and justify:** I can apply Topics 7.1–7.5 to solve AP-style equilibrium questions and justify my answers with calculations and reasoning.	7.3 Minimize threats and distractions through predictable structures and varied stimulation. 4.2 Ensure access to tools and assistive technologies. 8.4 Provide mastery-oriented feedback that emphasizes growth and improvement.	AP Classroom: Unit 7 checkpoint. AP Chemistry equations and constants sheet.

التقييم المرحلي للوحدة السابعة		**Graded formative:** College Board Unit 7 checkpoint (Topics 7.1–7.5) on AP Classroom	
Homework الواجبات			
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Session 4 is a graded formative assessment: the College Board Unit 7 checkpoint on AP Classroom, covering Topics 7.1–7.5. Students need a calculator and the AP equations and constants sheet. Sessions 1–3 build the skills for the checkpoint: writing Q and K expressions, calculating K with ICE tables, and interpreting the size of K.		

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Technology Integration دمج التكنولوجيا	AP Classroom (AP Daily videos, Topic Questions and the Unit 7 checkpoint) and Toddle for lesson materials.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics (exponents, rearranging equations and solving for unknowns in ICE tables) and Economics (how industry uses equilibrium to improve product yield).
External Assessment Integration دمج التقييمات الخارجية	AP Chemistry Exam (College Board): Course and Exam Description Unit 7; Unit 7 checkpoint on AP Classroom.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Self-management (preparing for and staying calm during the checkpoint), thinking (analysing data to calculate and interpret K) and communication (justifying answers in writing, as on the AP free-response section).
Standards المعايير	• **7.3 Reaction Quotient and Equilibrium Constant – TRA-7.A:** Represent the reaction quotient Qc or Qp for a reversible reaction, and the corresponding equilibrium expressions Kc = Qc or Kp = Qp. • **7.4 Calculating the Equilibrium Constant – TRA-7.B:** Calculate Kc or Kp based on experimental observations of concentrations or pressures at equilibrium. • **7.5 Magnitude of the Equilibrium Constant – TRA-7.C:** Explain the relationship between very large or very small values of K and the relative concentrations of chemical species at equilibrium.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Inquiry:** How can one number, K, tell us how far a reaction goes, and how do chemists use it to predict what is in the mixture at equilibrium? **TOK:** We never see K directly; we calculate it from measured concentrations. How certain can we be about a value we only know through other measurements?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Session 1 – 7.3 Reaction Quotient and Equilibrium Constant** • **MUST (DOK 1):** I can write the Kc expression for a balanced equation, with products over reactants raised to the power of their coefficients. • **SHOULD (DOK 2):** I can write Qc and Kp expressions, leave out pure solids and liquids, and explain that Q = K only at equilibrium. • **COULD (DOK 3):** I can justify which species are left out of Q and K in heterogeneous equilibria, and explain why Q changes as a reaction moves toward equilibrium. **Session 2 – 7.4 Calculating the Equilibrium Constant** • **MUST (DOK 1):** I can substitute equilibrium concentrations into a Kc expression to calculate K. • **SHOULD (DOK 2):** I can use an ICE table to find equilibrium concentrations and then calculate K. • **COULD (DOK 3):** I can calculate Kp from partial pressures and evaluate whether data from different trials give the same K at the same temperature. **Session 3 – 7.5 Magnitude of the Equilibrium Constant**

	• **MUST (DOK 1):** I can state that when K is much greater than 1 products are favored, and when K is much less than 1 reactants are favored. • **SHOULD (DOK 2):** I can rank several reactions by how far they proceed, using their K values. • **COULD (DOK 3):** I can justify, using K and concentration data or particle diagrams, why a reaction with a very large K is treated as going to completion. **Session 4 – Unit 7 Checkpoint (College Board)** • **MUST (DOK 1):** I can identify when a system is at equilibrium and write correct Q and K expressions. • **SHOULD (DOK 2):** I can calculate K from experimental data using an ICE table, with correct units and significant figures. • **COULD (DOK 3):** I can justify answers about the extent of a reaction and, after feedback, correct my errors and explain each misconception.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Arabic Language اللغة العربية **Area:المجال** Sustainable industry and energy in the UAE, linked to the UAE Net Zero by 2050 strategic initiative. **Implementation:التطبيق** • Sessions 1–2: students write Q and K expressions and calculate K for the reversible reaction that makes ammonia, as produced in Ruwais, using sample equilibrium data. • Session 3: students compare K values to explain why some industrial reactions give a high yield and others need controlled conditions to support UAE industry. • Arabic key terms: ثابت الاتزان (equilibrium constant), حاصل التفاعل (reaction quotient), التركيز (concentration), الضغط الجزئي (partial pressure), الاتزان الكيميائي (chemical equilibrium).